KHUSHI NAYAK

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PROFESSIONAL SUMMARY

Entry-level AI Engineer with a B.Tech in Artificial Intelligence & Machine Learning. Experience building Generative AI applications, Large Language Model (LLM) agents, Retrieval-Augmented Generation (RAG) pipelines, and deploying machine learning models via FastAPI REST APIs and Docker. Currently a Trainee AI & Machine Learning Engineer at Tedekstra, developing AI-driven backend services. Built agentic AI systems using Gemini and a deep learning model with 98% accuracy. Solved 250+ LeetCode problems (100+ Medium/Hard).

TECHNICAL SKILLS

Programming Languages: Python, SQL, C++, JavaScript

Generative AI & LLMs: Large Language Models (LLMs), Generative AI, Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Embeddings, Vector Databases, Gemini, LlamaIndex, Pinecone, Chatbots

Machine Learning & MLOps: Supervised/Unsupervised Learning, CNN, LSTM, Feature Engineering, Model Evaluation, Model Serving, Inference Pipelines, Exploratory Data Analysis (EDA)

Deployment, Cloud & Tools: FastAPI (Model Serving), Docker, REST APIs, Git, GitHub, Streamlit, Azure Databricks (Foundations)

Libraries & Frameworks: TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, React.js, Next.js, HTML, CSS

WORK EXPERIENCE

Trainee AI & Machine Learning Engineer, Tedekstra **June 2026 – Present**

- Develop AI-powered features for web applications using Python, machine learning, and Generative AI.
- Build AI-driven backend services and containerized REST APIs for model inference and intelligent automation.
- Integrate and optimize AI solutions to improve application functionality, reliability, and deployment performance.

Generative AI Intern, Prodigal AI Pvt. Ltd **September 2024 – December 2024**

- Built a Natural Language Processing (NLP) chatbot with intent classification and context-aware responses.
- Implemented a Retrieval-Augmented Generation (RAG) architecture with optimized embeddings to improve retrieval and response quality.

Data Analyst Intern, International Business Machines (IBM) **June 2024 – August 2024**

- Performed Exploratory Data Analysis (EDA) and machine learning modeling on water quality datasets, achieving 92% prediction accuracy.
- Compared Logistic Regression, SVM, and Random Forest models; selected the best model using precision and recall metrics.

PROJECTS

Deep Research AI – Autonomous Web Research Agent [GitHub] *Gemini, REST APIs, React*

- Designed an agentic AI pipeline using Gemini to decompose queries, evaluate source authority, and extract facts.
- Implemented secure REST APIs and a React dashboard with real-time research pipeline tracking.

AI Self-Healing Test Agent – Full-Stack & AI Project [GitHub] *React, FastAPI, Playwright, Vector Embeddings*

- Built an AI testing platform using React and FastAPI that converts plain text into Playwright browser tests.
- Implemented a vector embedding engine that automatically recovers broken UI selectors at runtime.

Packet Analyzer – Hybrid DPI + ML Traffic Classifier [GitHub] *C++, Python, FastAPI, Scikit-learn*

- Designed a hybrid C++ DPI and Random Forest model achieving 98.3% accuracy on encrypted traffic.
- Deployed a containerized FastAPI service (Docker) and dashboard to monitor encrypted network flows without payload decryption.

Seizure Prediction using SBM-MSTM Model [GitHub] *Deep Learning, EEG Signal Processing*

- Designed an SBM-MSTM deep learning model that achieved 98% accuracy on complex EEG signal datasets.
- Developed a decision-support tool to assist doctors in monitoring epilepsy patients.

EDUCATION

University School of Automation & Robotics, Delhi **2021 – 2025**
Bachelor of Technology (B.Tech) in Artificial Intelligence & Machine Learning

CERTIFICATIONS

- IBM SkillsBuild Summer Internship Program, Knowledge AI Agent

ADDITIONAL INFORMATION

Activities: Graphics Design Vice Lead, Association for Computing Machinery – Women (ACM-W)

Spoken Languages: English, Hindi